

Molecular Simulation

Computer Lab & Report Instructions

Flemingsberg Campus - T3/T4 - 06 February 2026 8.00-10.00

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Computer Lab - Group Task (2 persons in each group)

Simulating the dynamic behaviour of **40 Argon atoms** in **2 dimensions** for **50 ps** at a temperature of **300 K** in a box with a length of **4 nm** in a periodic boundary condition and a time step of **0.01 ps**, using the given Python script in the Canvas page.

Argon is a noble gas and has an atomic mass of **39.95 u.m.a**. The Argon atoms interact with van der Waals interactions, which we model using the Lennard-Jones **potential**:

$$V_{VDW}(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right]$$

The parameter ϵ is the well-depth, and the parameter σ is the distance at which the potential is zero. **Set σ equal to 0.34 nm and ϵ equal to 0.979 kJ/mol.**

Address the following questions in the report using the information above and the Python script. Ensure the text is clearly written and covers all the details. Max 1500 words.

Use captions to describe the content of each figure, labels for axis definition, and legends for set definition. Report values together with units.

NOTE: Keep all the versions of the Python script in case they are requested during the corrections.

The report should be uploaded to Canvas as soon as possible, and no later than **18th February 2026**. **Only one resubmission is possible**. Only if the issue is addressed properly in the first resubmission will you pass the lab.

Points to address in the report

a) The atoms interact with van der Waals interactions, which we model using the Lennard-Jones potential. As explained earlier, the parameter ϵ is the well depth. Perform a simulation for a new particle-type, having a Lennard-Jones potential with a depth of 1.5 kJ/mol (keep the initial σ value) and compare the

results with the simulations you have already performed for Argon atoms. In particular:

Draw the potential energy curve as a function of the distance and compare the shapes. Add a plot with both curves in it and explain what has changed.

Compare the total and potential energies you got from Argon and new particle-type simulations. Support your analysis with suitable graphs.

b) Total energy ($E_{\text{tot}} = E_{\text{pot}} + E_{\text{kin}}$) in the simulation should be conserved, so monitoring this during the simulation is good practice. Describe what happens to the total energy of the Argon simulation if you use a time step larger than 0.01 ps. What is the largest time step that one can use to simulate the Argon atoms and keep energy conservation? Support your results with simulation outputs.

c) Describe how temperature and mass affect the kinetic energy. To achieve this, run simulations with different values of temperature and mass. Tips: change only one value in each simulation and not more than 10%